1. Cooper and McGillem book

   5-1.1, 5-3.2,

2. A random process has sample function in the following form

   $$X(t) = A\left[\cos(2\pi ft + \Theta) + \sin(2\pi ft + \Theta)\right]$$

   Where A is a random variable with mean 2 and variance 4, and Θ is uniformly distributed in $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$. Find

   (a) The mean function of X(t)

   (b) The variance function of X(t)

   (c) The autocorrelation function of X(t)

   (d) The autocovariance function of X(t)

3. A discrete-time random process is defined by $X(n) = Y^n$, where Y is uniformly distributed in (0, 1).

   (a) Find the CDF of X(n)

   (b) Find the mean function and variance function of X(n)

   (c) Find the autocorrelation function $R_X(m,n) = E[X(m)X(n)]$

4. The random process Z(t) is defined by $Z(t) = 2Xt - Y$, where X and Y are two correlated random variables with means m_X, m_Y, variances σ_X^2, σ_Y^2, and correlation coefficient ρ. Find

   (a) The mean function of Z(t)

   (b) The auto-covariance function of Z(t)